SWAGAT NEUPANE

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SUMMARY

Computer Science graduate focused on applied machine learning and software engineering. Owned the ML forecasting component of a team capstone, from data cleaning and feature engineering to chronological evaluation against a baseline and verified model loading. Strongest in Python, pandas, and scikit-learn, with a habit of honest evaluation and reproducible, tested pipelines.

EDUCATION

San Francisco Bay University *Fremont, CA*
Bachelor of Science in Computer Science *August 2026*

TECHNICAL SKILLS

- **Languages:** Python, SQL (MySQL), JavaScript, HTML, CSS, Bash
- **ML & Data:** pandas, NumPy, scikit-learn (Random Forest, Gradient Boosting), feature engineering, time-based evaluation, MAE/WAPE, Matplotlib, Seaborn, Jupyter
- **Engineering:** Git/GitHub, pytest, joblib model versioning, Linux, GitHub Pages

PROJECTS

RestroPilot | *ML Developer, Team Capstone* *Python, pandas, scikit-learn, pytest, joblib*

- Owned the machine-learning component of a team restaurant-operations platform and built its original reproducible forecasting prototype with pytest coverage.
- Cleaned and reconciled 194K order lines (33K orders, 550 catalogue items) from four CSV sources across two restaurants, keeping source provenance, flagging ambiguous duplicates, and treating missing dates as unknown instead of zero sales.
- Trained a Random Forest to predict daily item quantities using 1/7/14/28-day lags and 7/28-day rolling means, with strict chronological train/validation/test splits and tuning on validation only.
- Cut test MAE by about 25% against a same-item previous-week baseline (4.17 vs. 5.57 units per scored item-day) on held-out historical data, reporting per-item error and excluded-day coverage.
- Built versioned model loading with SHA-256 artifact checks and a historical inference service feeding the app's forecast page.

Bay Area Environmental Data Analysis *Python, pandas, scikit-learn, Seaborn*

- Prepared county-level pollution, socioeconomic, and asthma data, handling missing values and comparing counties.
- Built correlation visualizations and ran exploratory Random Forest and Gradient Boosting regression experiments.

Personal Portfolio Website *HTML, CSS, JavaScript, GitHub Pages*

- Built and hosted a static portfolio with project case studies, a keyboard-accessible calculator, and a filterable showcase page, with SVG illustrations and reduced-motion support.

Driving School Database *MySQL*

- Designed a relational schema for students, instructors, schedules, and registration, with queries for scheduling.

Command-Line Tic-Tac-Toe *Bash, Linux*

- Wrote a two-player terminal game with game logic and input validation.

LEADERSHIP & ACTIVITIES

- House Captain and Class Leader, Uniglobe Higher Secondary School: led a class of 10 students and helped organize a chess competition and an academic quiz; IT Club member.
- Volunteer, Bagmati River cleanup campaign and a community book donation program.